MIL-V-17737F(SH)
4 December 1981
SUPERSEDING
MIL-V-17737E(SHIPS)
21 January 1976
(See 6.7)

MILITARY SPECIFICATION

VALVES, BOILER BLOW, SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 1-1/2 inch blow valves (hard seat and disc) for bottom blow, guarding, and header blow service at nominal working gage pressure of 600 or 1500 pounds per square inch (lb/in^2).

1.2 Classification. Blow valves of the straightway or angle pattern shall be of the following types and classes, as specified (see 6.2.1):

Type I - Flanged blow.
Class 1 - 600 lb/in^2 service.
Class 2 - 1500 lb/in^2 service.
Type II - Flanged blow, with valve stem adaptor and mating, unattached tee wrench.

Class 1 - 600 lb/in^2 service.
Class 2 - 1500 lb/in^2 service.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-763 - Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-V-3 - Valves, Fittings and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-R-196 - Repair Parts, Accessories and Kits, Mechanical, Packaging of.
MIL-S-862 - Steel Bars, Corrosion Resisting, and Steel Billets, Corrosion Resisting; Reforging Application.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-S-1222 - Studs, Bolts, Hex Cap Screws, and Nuts.
MIL-P-15024 - Plates, Tags, and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-R-17131 - Rods and Powders, Welding, Surfacing.
MIL-P-17303 - Packing Materials, Plastic Metallic and Plastic Nonmetallic.
MIL-G-21032 - Gaskets, Metallic-Asbestos, Spiral Wound.
MIL-P-24503 - Packing Material, Graphic, Corrugated Ribbon and Preformed Ring.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.

2.1.2 Government drawings. The following Government drawing forms a part of this specification to the extent specified herein.

DRAWING

NAVAL SEA SYSTEMS COMMAND

NAVSHIPS 5000-S4824-1385620 - Handwheels for Valves.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B16.5 - Steel Pipe Flanges and Flanged Fittings (DoD adopted).
B16.34 - Steel Valves.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 105 - Forgings, Carbon Steel, for Piping Components.
A 182 - Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service (DoD adopted).
A 314 - Stainless and Heat-Resisting Steel Billets and Bars for Forging.
A 582 - Free-Machining Stainless and Heat-Resisting Steel Wire (DoD adopted).
B 564 - Nickel Alloy Forgings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials. Materials shall be as specified hereinafter or shall be of equal or superior properties for the intended pressures and temperatures, and shall be selected so as to prevent galling, seizing, or excessive wear on operating parts. Clearances shall be such as to prevent interferences due to thermal expansion. The valves shall be used for handling and continuous use with boiler water. Boiler water for the purpose of this specification is water having 500 parts per million (p/m) undissolved boiler sludge and 1150 p/m dissolved solids of which 500 p/m may be chlorides. The suitability of the material of all other parts not specified in this specification and the substitution of any of the specified material for a better material shall be subject to review by the command or agency concerned and then its suitability shall be demonstrated by satisfactory performance of the valves during the tests specified in 4.3.

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3.2.1 Bodies, bonnets, and yoke retaining ring. Bodies, bonnets, and yoke retaining ring shall be of forged carbon steel conforming to ASTM A 105, or of forged alloy steel conforming to ASTM A 182, grade F22.

3.2.2 Valve stems. Valve stems shall be of corrosion-resisting steel, conforming to A416 stainless steel composition and heat-treated to 24-26 Brinell hardness in accordance with MIL-S-862, ASTM A 314, or ASTM A 582.

3.2.3 Bolts, studs, cap screws, and nuts. Bolts, studs, cap screws, and nuts shall be alloy steel and shall conform to MIL-S-1222.

3.2.4 Handwheels. Handwheels shall be in accordance with drawing 5000-S4824-1385620. Handwheel material shall be aluminum alloy. Valves shall close by clockwise rotation of the handwheel when facing the handwheel.

3.2.5 Hard seat (disc and seat). Valve seats and discs shall be hard-faced by depositing a layer of cobalt-chromium alloy conforming to type MIL-RCoCr-A-1 or MIL-RCoCr-A-2 of MIL-R-17131 with a finished thickness of hard-faced surface of not less than 3/32 inch or shall be solid cobalt-chromium (stellite) discs and seats.

3.2.5.1 Valve discs. Valve discs shall conform to QQ-S-763 or to grade F11 of ASTM A 182, with seating area faced with cobalt-chromium composition (see 3.2.5) deposited by welding to insure positive bond between it and the disc, or shall be a solid cobalt-chromium composition (stellite). The valve disc shall be of the swivel attachment type. The disc shall incorporate a blow deflector feature which successfully completes the 200 cycle performance test (see 4.6.2) without seat damage (seat wear and wire drawing) and without seat leakage in excess of that specified in 3.3.8. A hardened washer (500 Brinell minimum) or pad shall be inserted between the end of the stem and the disc, or a pad of stellite shall be buttered to the inside of the disc at a point of contact, to prevent galling.

3.2.5.2 Valve seats. Valve seats shall be an integral part of the valve body or may consist of a tubular insert welded into the body, with sufficient body material provided to permit the fitting of renewable seats, if necessary. The seating surfaces shall be faced with cobalt-chromium composition (see 3.2.5), or shall be a solid cobalt-chromium composition (stellite).

3.2.5.3 Stuffing boxes and glands. The packing gland shall conform to ASTM B 564 (UNS N04400) or to ASTM A 105. The stuffing box gland shall be accessible without removal of stem or valve from the piping system. Adjustment of the stuffing box gland shall insure tightness of the stuffing box under all operating conditions. Valve stem packing glands shall be of ample depths, to hold at least five turns of packing, to insure tightness against leakage under severe conditions.

3.2.5.4 Packing and gaskets. Packing shall conform to class II, type C (850 degrees Fahrenheit (^{o}F) maximum) or type E (1050°F maximum) of MIL-P-17303 as applicable, or MIL-P-24503. Gaskets shall be spiral wound in accordance with type II of MIL-G-21032.

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3.2.6 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable, without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification, unless otherwise specifically specified.

3.3 General design. Blow valves shall withstand a nominal working gage pressure of 600 or 1500 lb/in^2 to insure positive tightness under severe service. The valves shall be reliable and maintained in service at their intended nominal pressure for long periods of time, without adjustments or overhauls. Valves shall be designed in accordance with ANSI B16.34.

3.3.1 Flange dimensions. The valve flange dimensions shall be in accordance with ANSI B16.5. For class 1 valves, the flange face-to-face dimension shall be 12 inches ± 1/16 inch for straightway pattern, and the inlet flange-face to centerline and centerline to outlet face shall be 4-3/4 inches for the angle pattern. For class 2 valves, the flange face-to-face dimension shall be 15-1/2 inches ± 1/16 inch for straightway valves, and the inlet flange-face to centerline and centerline to outlet face shall be 6-1/2 inches for angle valves.

3.3.1.1 Flange finish. The gasket-contact surfaces of flanges shall be finished with a circular lay (concentric or phonographic) having a roughness not exceeding 500 roughness height rating (RHR) produced by machining cuts of 0.003 inch maximum depth but not less than 40 cuts per inch of face width. Finish is to be entered on applicable drawings (see 3.7).

3.3.1.2 Assembly dimensional tolerances. The flanged end connections shall be perpendicular to the centerline of the valve in both the X and Y axis. The maximum allowable tolerance is plus or minus 1/16 inch. The flanges shall be alined with the top two bolt holes parallel to the horizontal centerline of the flange. The maximum allowable flange rotation is plus or minus 1/16 inch.

3.3.2 Flow characteristics. Valves shall have an unrestricted area through the valve, with the valve in full open position, and through all body passages, not less than 80 percent of the area of the pipe to which the valves will be attached. A high lift feature, permitting withdrawal of the valve disc into the bonnet for protection during operation, at least equal to full flow passage, shall be incorporated.

3.3.3 Valve bonnets. Valve bonnet shall be flanged and secured by bolt or stud and nut, or cap screw arrangement, or incorporate a nongasketed pressure seal arrangement. The valve bonnet alinement shall be obtained by body guiding, that is a close tolerance fit between bonnet and valve body. The manufacturer shall demonstrate bonnet design to ensure ease of bonnet removal for valve repair. Flange face designs shall assure proper compression and limit the compression on flange gasket when flanges are drawn up metal-to-metal.

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3.3.4 Yokes. Valves shall be of the outside screw and yoke construction. Unless otherwise specified (see 6.2.1), yokes shall be integral with the bonnets or Acme screw threaded to body with a bolted or cap screw securing device. The yoke shall be fitted with a bushing, threaded to fit the stem.

3.3.5 Stems. Valves shall have rotating stems with Acme threads, self-locking, and stems shall turn clockwise to close the valve.

3.3.5.1 Stem angle. The angle between the centerline of the stem and the centerline through the outlet flange shall be within the range of 45 degrees to 50 degrees.

3.3.6 Valve passage. The passage through the valves shall give a smooth flow through the valves without any abrupt changes in the direction of flow, in order to minimize the possibility of grease, scale, and sludge from accumulating in the valve.

3.3.7 Handwheel operation. Valves shall be relatively easy to open and close while under their intended nominal working pressure. Maximum allowable operating forces for valve handwheels shall be as specified in table I. Valves for 600 lb/in^2 service shall have a handwheel not less than 8 iches in diameter; valves for 1500 lb/in^2 service not less than 11 inches in diameter.

TABLE I. Maximum allowable operating forces for handwheels or levers.

Handwheel diameter or lever length	Total tangential force on rim of handwheel or end of lever	Equivalent torque
(Inches)	(Pounds)	(Maximum tolerance plus or minus 5 foot-pounds)
8 (5-inch tee handle length)	124	41
9 (maximum for class 1 valves)	128	48
10 (6-inch tee handle)	130	54
11	132	61
12 (maximum for class 2 valves: 7-inch tee handle length)	136	68

3.3.8 Seat leakage. Unless otherwise specified (see 6.2.1), allowable seat leakage is considered to be leakage of water not in excess of 10 milliliters (mL) per hour per inch of nominal pipe size, when tested in accordance with 4.6.2 and 4.6.4.2.

3.3.9 Coldness. Valves shall be cold tested as specified in 4.6.1.

3.3.10 Hydrostatic. Valves shall be hydrostatic tested as specified in 4.6.4.

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3.3.11 Welding and fabrication. Welding shall be in accordance with MIL-STD-278. Fabricated assemblies shall be stress relieved as units, prior to finish machining. The stress relieving shall be done in accordance with MIL-STD-278.

3.3.12 Shock and vibration.

3.3.12.1 Shock. The valve shall not be damaged or malfunction when tested as specified in 4.6.5.1.

3.3.12.2 Vibration. Valves shall not be adversely affected by externally imposed vibrations of up to and including 50 hertz (Hz). Valves shall withstand the vibration tests as specified in 4.6.5.2.

3.3.13 Interchangeability. Similar parts of corresponding valves furnished on the same contract or order or built to the same drawings, shall be strictly interchangeable without the necessity of further machining or handfitting of any kind. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.4 Type I - flanged blow valve.

3.4.1 Handwheels. Type I valves shall be provided with handwheels (see 3.2.4). The handwheels shall be secured to the stem by means of a hexagonal nut of suitable material, threaded to the stem. The outside diameter of the handwheel shall comply with 3.3.7.

3.5 Type II - flanged blow valve, with adaptor and tee wrench.

3.5.1 Valve stem connection. Type II valve shall be provided with valve stem tee handle adaptor and companion tee handle (wrench).

3.5.2 Tee handle and adaptor. A tee handle (wrench) with companion stem adaptor shall be furnished (in lieu of handwheel) with each type II valve with tee handle material and size conforming to 3.2.4 and 3.3.7, respectively. Adaptor shall be short steel lug, 3 inches maximum length, which will prevent manual operation of valve by lug itself and allow for connection to tee handle (wrench) for valve operation. Tee handle will be connected to adaptor only during valve operation.

3.6 Body marking. Blow valves shall be plainly marked by the contractor. The marking may be stamped on the body, cast on the body, or stamped on an identification plate permanently secured thereto, conforming to type C of MIL-P-15024 and the normal service requirements of MIL-P-15024/5. An arrow shall be stamped or cast on the body to designate the direction of flow. Identification markings shall contain the following information:

(a) Name of identifying trademark of manufacturer and designating number or symbol.
(b) The nominal size and nominal operating pressure for which the valve is intended, preferably cast on valve body, as for example, 1-1/2 inches - 600 or 1-1/2 inches - 1500.

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(c) When applicable, a "no backseat" warning.
(d) Component identification number (CID).

3.7 Drawings. When the inherent design of a valve is such that it cannot conform to 3.3.1 and following conclusion of a satisfactory test (see 4.3), the contractor shall furnish the command or agency concerned an outline drawing (tracing) of the valve, showing all limiting external dimension. Valves of this general type, which may subsequently be furnished for Military use, shall then conform to this dimensional standard drawing.

3.7.1 The contractor shall prepare drawings, including a list of materials, in accordance with the data ordering document included in the contract or order (see 6.2.2 and 6.4).

3.8 Tools. Tools shall be furnished as specified (see 6.2.1), the quantity depending upon the number of boilers installed in the ship. These tools will normally consist of special tools and wrenches for the proper maintenance and operation of the types of valves. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service Management Area (DCASMA)).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system in accordance with MIL-I-45208.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. One class 2 valve of either type shall be examined and tested at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). First article inspection shall consist of the examination specified in 4.5 and the tests specified in 4.6.1 through 4.6.5.2. A contractor whose class 2 valve has satisfactorily passed the first article inspection need not submit class 1 valves for first article inspection.

4.3.1 Test reports. The contractor shall prepare a first article test report in accordance with the data ordering document included in the contract or order (see 6.2.2).

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination in 4.5 and tests specified in 4.6.3 through 4.6.4.3.

4.4.1 Lot. Valves of the same type and class presented for delivery at one time shall be considered a lot.

4.4.2 Sampling for examination. A random sample of valves shall be selected from each lot in accordance with table II for the examination specified in 4.5 through 4.5.1.2.

TABLE II. Sampling for examination.

Number of valves in lot	Number of valves in sample	Acceptance number (defectives)
15 and under	7	0
16 to 25	10	0
26 to 40	13	0
41 to 65	16	0
66 to 110	21	0

4.4.3 Sampling for tests. Each valve in the lot shall be tested as specified in 4.6.3 through 4.6.4.3.

4.5 Examination. Each of the sample valves selected in accordance with 4.4.2 shall be visually and dimensionally examined to determine conformance with the requirements of this specification as to material, workmanship, and dimensions specified on the drawings, and for such modifications as may be required as a result of the first article inspection (see 4.3). Material examination shall be as specified in 4.5.1 through 4.5.1.2.

4.5.1 Pressure containing parts shall be examined to determine conformance with the applicable material specification.

4.5.1.1 The DCASMA shall normally accept certification that nonpressure containing parts conform to their applicable material specification, providing the certification contains actual test results. Certification representing nonmetallic stocks more than one year old are not acceptable.

4.5.1.2 When there is a question relative to material identification or quality of a lot, the DCASMA may request a chemical analysis and mechanical tests as he deems necessary to establish complete identity and overall quality.

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4.6 Tests.

4.6.1 Cold tests. Cold tests shall be performed as follows:

(a) Operation - 1 to 10 cycles.
(b) Hydrostatic - shell and seat tightness (see 4.6.4.1 and 4.6.4.2).

4.6.2 Performance test. Blow down tests shall be conducted with boiler water at the pressure and temperature for which the valve is rated, blowing down to atmospheric pressure for 200 cycles. The blow down cycle shall consist of opening the valve rapidly to the full open position and immediately closing the valve quickly. The valve shall remain closed for at least 5 minutes between cycles. After each 20 closures and after 200 blow down cycles, the valve shall be tight. Upon completion of 200 blow down cycles, four securing and warm-up trials shall be made to determine that the valve remains tight, after the valve has been closed, during and after cooling to room temperature (65°F to 95°F) and during warm-up to saturated temperature conditions. The operating torque shall be taken and recorded periodically to assure that a man can open and close a valve whether the valve was closed at room temperature or at the operating temperature of the boiler. The four warm-up tests shall be in accordance with the following schedule:

(a) Two tests at a boiler warm-up rate of 2 hours.
(b) Two tests at a boiler warm-up rate of 30 minutes.
(c) Valve seat leakage shall not exceed the requirements specified in 3.3.8 during warm-up, following cold closure, when above tests are run and closure torque are as specified in table I.

4.6.3 Stellited seating. Stellited seating surfaces shall be liquid penetrant inspected in accordance with MIL-STD-271, after rough machining, and shall be free of cracks and crack-like defects.

4.6.4 Hydrostatic tests. Each valve shall be subjected to the tests specified in 4.6.4.1 for strength and porosity, and tests specified in 4.6.4.2 for tightness. The water temperature shall not exceed 100°F.

4.6.4.1 Shell test. Valves shall be subjected to pressures specified in table III for a duration of 10 minutes. Any weeping, leakage, or permanent deformation shall be cause for rejection.

TABLE III. Shell test pressures for strength and porosity.

Ratings ANSI	Gage pressures
Pounds	lb/in^2
600	2175
1500	5400

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4.6.4.2 Seat tests (tightness). Valves shall be pressurized in the direction tending to open the valve and examined for seat tightness. Test pressures shall be as specified in table IV. Duration of the test shall be 3 minutes minimum. There shall be no visible signs of leakage. Should any visible leakage be seen, the test shall be continued for a length of time to accurately determine the rate of leakage. Leakage shall not exceed that specified in 3.3.8. The maximum allowable force to seat valves during the test shall not exceed the value listed in table I.

TABLE IV. Test pressures for seat tightness.

Ratings ANSI	Gage pressures
Pounds	lb/in^2
600	900
1500	2250

4.6.4.3 Equivalent torque. Test for compliance with table I shall be made using the 100 percent hydrostatic test condition (see 3.3.7).

4.6.5 Shock and vibration.

4.6.5.1 Shock. One valve of each new design shall be tested in accordance with grade A, class 1, type A of MIL-S-901.

4.6.5.2 Vibration. Valve(s) shall be subjected to the vibration tests in accordance with type I of MIL-STD-167-1.

4.7 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6).

5.1 Preservation-packaging, packing, and marking. Valves shall be individually preserved-packaged level A or C, packed level A, B, or C, and marked in accordance with MIL-V-3, as specified (see 6.2.1).

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5.2 Tools. Unless otherwise specified (see 6.2.1), tools shall be preserved-packaged, packed and marked in accordance with the levels of PPP-P-40 or MIL-R-196, as applicable for the intended use and destination as follows:

	Levels	
	Preservation-packaging	Packing
On board	A	C
Stock	A	B
Immediate use	C	C

5.3 Cushioning, dunnage, and wrapping materials.

5.3.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, or dunnage, is prohibited for materials destined for shipboard installation or stowage.

5.3.2 Level C preservation-packaging and packing. When loose-fill type materials are used for packaging and packing applications such as cushioning, filler, and dunnage, all containers (unit, intermediate, and shipping) shall be marked or labelled with the following information:

"CAUTION

Contents cushioned with loose-fill material shall not be taken on board ship. Remove and discard loose-fill material before shipboard stowage. If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material."

5.3.3 Cushioning, filler, dunnage and wrapping materials selected, whenever available, shall exhibit improved performance for resistance to fire.

5.4 Data (see 3.7 and 6.2.2). Drawings shall be prepared for shipment in accordance with the applicable specification. Other data, reports, plans and so forth, shall be prepared for shipment in accordance with the level C requirements of MIL-V-3.

6. NOTES

6.1 Intended use. The boiler blow valves covered by this specification are intended for use in freeing a boiler of internal accumulations of grease, scale, and sludge and in reducing the salinity of a steaming boiler.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2).
(c) If first article inspection is required (see 3.1).
(d) Yoke construction (see 3.3.4).
(e) Allowable seat leakage (see 3.3.8).
(f) Tools required (see 3.8).
(g) Selection of applicable levels of preservation-packaging and packing required (see 5.1).
(h) If tools are to be packaged other than specified (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.7.1	Drawings, engineering and associated lists	DI-E-7031	Level 2 Design activity designation - Government Drawing number - Government Delivery of hard copy - Contracting activity
4.3.1	First article inspection report	DI-T-4902	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 A list of material of valves shall be prepared by the contractor for the Naval Sea Systems Command, Washington, DC 20362 and the Naval Ship Systems Engineering Station, Philadelphia, PA 19113.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
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